

SPACE AND TIME

34–37 Radiation

38–39 Gravity, motion, and orbits

Expanding space 44–45

The family of the Sun 102–103

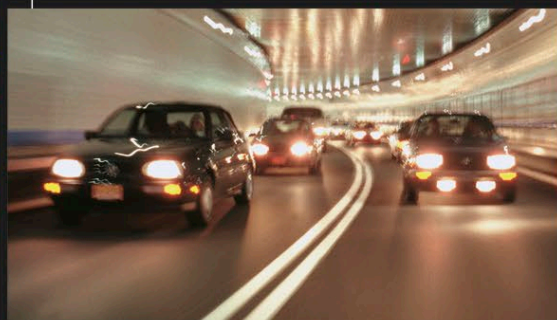
MOST PEOPLE SHARE SOME COMMON-SENSE NOTIONS about the world. One is that time passes at the same rate for everyone. Another is that the length of a rigid object does not change. In fact, such ideas, which once formed a bedrock for the laws of physics, are an illusion: they apply only to the restricted range

of situations with which people are most familiar. In fact, time and space are not absolute, but stretch and warp depending on relative viewpoint. What is more, the presence of matter distorts both space and time to produce the force of gravity.

PROBLEMS IN NEWTON'S UNIVERSE

Problems with the Newtonian view of space and time (see p.38) first surfaced toward the end of the 19th century. Up to that time, scientists assumed that the positions and motions of objects in space should all be measurable relative to some nonmoving, absolute “frame of reference,” which they thought was filled with an invisible medium called “the ether.” However, in 1887, an experiment to measure Earth’s motion through this ether, by detecting variations in the velocity of light sent through it in different directions, produced some unexpected results.

First, it failed to confirm the existence of the ether. Second, it indicated that light always travels at the same speed relative to an observer, whatever that observer’s own movements. This finding suggested that light does not follow the same rules of relative motion that govern everyday objects such as cars and bullets. If a person were to chase a bullet at half of the bullet’s speed, the rate at which the bullet moved away from him or her would halve. However, if a person were to chase a light wave at half the speed of light, the wave would continue moving away from him or her at exactly the same velocity.



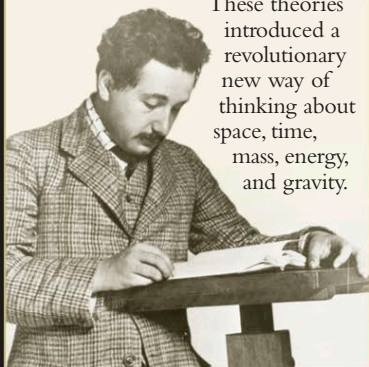
CONSTANT SPEED OF LIGHT

Light leaves both the ceiling lights and the headlights of the moving cars at the same speed relative to its source. Paradoxically, light from both sources reaches an observer standing in the tunnel at, again, exactly the same speed.

ALBERT EINSTEIN

The work of German-born mathematician and physicist Albert Einstein (1879–1955) made him the most famous scientist of the 20th century. Although he won the Nobel Prize for his work on the particlelike properties of light (see p.34), he is more famous for his theories of special relativity (1905) and general relativity (1915).

These theories introduced a revolutionary new way of thinking about space, time, mass, energy, and gravity.



SPECIAL RELATIVITY

In 1905, Albert Einstein rejected the idea that there is any absolute or “preferred” frame of reference in the universe. In other words, everything is relative. He also rejected the idea that time is absolute, suggesting that it need not pass at the same rate everywhere. To replace the old ideas, he formed the special theory of relativity, called “special” because it is restricted to frames of reference in constant, unchanging motion (because they are not being accelerated by a force, see p.38). He based the entire theory on two principles. The first principle, called the principle of relativity, states that the same laws of physics apply equally in all constantly moving frames of reference. The second principle states that the speed of light is constant and independent of the motion of the observer or source of light. Einstein recognized that this second principle conflicts with accepted notions about how velocities add together, and further, that combining it with the first principle seems to lead to perplexing, nonintuitive results. He perceived, however, that human intuition about time and space could be wrong.

MOVING FRAMES OF REFERENCE

Here, we see two travelers—effectively two moving reference frames—passing each other. Each tosses a ball up. By the principle of relativity, the laws of physics apply in each reference frame, so each traveler observes the behavior of the two balls as predicted by those laws. Although the two travelers see different motions for each ball, neither traveler’s point of view is superior to the other’s—both are equally valid, and there is no preferred frame of reference.

direction of
Observer 2's
motion

path of
Observer 1's
ball, as seen
by Observer 2

Observer 1

path of Observer
1's ball, as seen
by Observer 1

VIEWPOINT ONE

From Observer 1's point of view, the green ball within his or her own frame of reference travels up and down. If Observer 1 looks across at the red ball in a frame of reference in relative motion, the ball seems to follow an arc.

direction of
Observer 1's
motion